CAGE-Fake: Article Trustworthiness Audit Report

Calibration-Aware Gating and Evidence-Grounded Reasoning for Trustworthy Fake News Triage

LEGAL & METHODOLOGICAL DISCLAIMER: *A trained fake-news classifier predicts linguistic, stylometric, and semantic patterns learned from benchmark datasets; it does **NOT** independently verify whether a news story is factually true. URL extraction, model pattern recognition, and external fact verification remain strictly separate processes. High uncertainty cases are flagged for professional human editorial review.*

Source / URL: https://example.com/senate-clean-energy
Title: Senate Approves Major Clean Energy Infrastructure Bill
Word Count: 450 words | **Date:** 2026-10-20 20:03:33

VERDICT: REAL

Calibrated Confidence: 94.2% (Threshold tau = 80.0%)
Base Disagreement (std): 0.035 (Threshold delta = 0.20)
Gating Trigger: None

1. Multi-Model Disagreement Analysis (5 Models)

Model Name	P(FAKE)	Model Vote	Confidence
Random Forest	8.0%	REAL	92.0%
XGBoost	5.0%	REAL	95.0%
LightGBM	6.0%	REAL	94.0%
Logistic Regression	9.0%	REAL	91.0%
CAGE-Fake Stack	5.8%	REAL	94.2%

Consensus Status: Unanimous Consensus
Base Disagreement (4-base std): 0.035 (Delta = 0.20)
5-Model Consensus Spread: 0.015
Vote Distribution: 0 FAKE vs 5 REAL
Diagnostic: All 5 models exhibit nearly identical probability outputs.

2. News Credibility Fingerprint (Linguistic & Stylistic Scorecard)

Credibility Dimension	Score (0-100)	Journalistic Assessment
Sensationalism Discipline	92.0/100	Excellent
Source Attribution	85.0/100	Excellent
Epistemic Grounding	88.0/100	Excellent
Punctuation Discipline	95.0/100	Excellent
Readability Structure	80.0/100	Excellent
Lexical Diversity	78.0/100	Adequate

Overall Credibility Index: 86.8/100
Credibility Tier: A (Restrained / Professional Copy)

Scores are normalized against verified newsroom style guides: high attribution, balanced syntax, and low sensationalism maximize the index.

3. Transparent Reasoning & Counterfactual Sensitivity

Diagnostic Explanation:
The article presents well-attributed statements with high inter-model consensus.

Counterfactual Simulation: Article employs factual, restrained syntax.
Original P(FAKE): 5.8% | **Transformed P(FAKE):** 5.8% **(Signed Shift: +0.0 percentage points)**

4. Local Signal Attributions (Meta-Ensemble Impacts)

Top FAKE-Driving Indicators	Weight / Impact	Top REAL-Driving Anchors	Weight / Impact
Lexical Diversity	+0.010	Journalistic Source Attribution Density	-0.250

5. Empirical Benchmark Model Comparison (Held-out Test Split)

Model	Accuracy	Macro-F1	ECE	Coverage	Selective Acc
Random Forest	0.8900	0.8900	0.0700	100.0%	0.8900
CAGE-Fake (Calibrated + Gated)	**0.9300**	**0.9300**	**0.0300**	**82.0%**	**0.9700**